More generally, if Q and K are arbitrary $d \times d$ matrices, then the same result also holds for the Cauchy problem for (2.3) with $V = I_d$ (or the natural analogue of (USA) with these parameters).

Proof. We focus on the dynamics (SA), but the proof is in fact identical in the case of (USA).

For $\alpha \in [0,1)$, we say that a set formed from n points $z_1, \dots, z_n \in (\mathbb{S}^{d-1})^n$ is *α–clustered* if for any $i, j \in [n]$, $\langle z_i, z_j \rangle > \alpha$ holds. Observe that if $\{z_1, \dots, z_n\}$ is α–clustered for some $\alpha \geqslant 0$, then the solution to the Cauchy problem for (SA) (for arbitrary $\beta \geqslant 0$) with this sequence as initial condition converges to a single cluster, since $w = z_1$ satisfies the assumption in Lemma 6.4.

Now, for any integer $m \geqslant 1$, we denote by $\mathcal{S}_0^m \subset \mathcal{S}_0$ the set of initial sequences $x_1(0), \dots, x_n(0)$ in $(\mathbb{S}^{d-1})^n$ for which the solution $(x_i^0(\cdot))_{i\in[n]}$ to the associated Cauchy problem for (4.1) is $\frac{3}{4}$–clustered at time $t = m$, namely

$$\langle x_i^0(m), x_j^0(m) \rangle > \frac{3}{4} \tag{4.2}$$

holds for all $i, j \in [n]$. We see that $\mathcal{S}_0^m$ is an open set for any integer $m \geqslant 1$. Moreover, $\mathcal{S}_0^m \subset \mathcal{S}_0^{m+1}$ according to the proof of Lemma 6.4, and $\bigcup_{m=1}^{+\infty} \mathcal{S}_0^m = \mathcal{S}_0$. This implies that

$$\lim_{m\to+\infty} \mathbb{P}(\mathcal{S}_0^m) = 1. \tag{4.3}$$

We now show that the solution to (SA) is near that of (4.1), starting from the same initial condition, when β is small. Using the Duhamel formula, we find

$$\begin{aligned} x_i^\beta(t) - x_i^0(t) &= \int_0^t \sum_{j=1}^n \left(\frac{e^{\beta\langle Q x_i^\beta(s), K x_j^\beta(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle Q x_i^\beta(s), K x_k^\beta(s)\rangle}} \right) \mathbf{P}^\perp_{x_i^\beta(s)}(x_j^\beta(s))\,\mathrm{d}s \\ &\quad - \int_0^t \frac{1}{n} \sum_{j=1}^n \mathbf{P}^\perp_{x_i^0(s)}(x_j^0(s))\,\mathrm{d}s \\ &= \int_0^t \sum_{j=1}^n \left(\frac{1}{n} + O\left(\frac{\beta}{n}\right) \right) \mathbf{P}^\perp_{x_i^\beta(s)}(x_j^\beta(s))\,\mathrm{d}s \\ &\quad - \int_0^t \frac{1}{n} \sum_{j=1}^n \mathbf{P}^\perp_{x_i^0(s)}(x_j^0(s))\,\mathrm{d}s, \end{aligned}$$

where we used that all particles lie on $\mathbb{S}^{d-1}$ for all times. Employing Grönwall, we deduce

$$\left\| x_i^\beta(t) - x_i^0(t) \right\| \leqslant O(\beta) e^{3t} \tag{4.4}$$

for all $t \geqslant 0$, $\beta \geqslant 0$ and $i \in [n]$. Due to (4.4), there exists some $\beta_m > 0$ such that for any $\beta \in [0, \beta_m]$,

$$\left\| x_i^\beta(m) - x_i^0(m) \right\| \leqslant \frac{1}{8}. \tag{4.5}$$

For this to hold, we clearly need $\beta_m \to 0$ as $m \to +\infty$. Combining (4.2) and (4.5), we gather that for any initial condition in $\mathcal{S}_0^m$, the solution $(x_i^\beta(\cdot))_{i\in[n]}$ to the corresponding Cauchy problem for (SA) is $\frac{1}{2}$–clustered at time $t = m$, namely satisfies

$$\langle x_i^\beta(m), x_j^\beta(m) \rangle > \frac{1}{2}$$

for all $i, j \in [n]$ and $\beta \in [0, \beta_m]$. Thus $\mathcal{S}_0^m \subset \mathcal{S}_\beta$ for any $\beta \in [0, \beta_m]$ by virtue of Lemma 6.4, which together with (4.3) concludes the proof. □

In the specific case where $Q^\top K = I_d$, we can in fact significantly sharpen Theorem 4.2 by relying on the gradient flow structure evoked in Section 3.4. Namely, we can show the following.

Theorem 4.3. *Fix $d, n \geqslant 2$. There exists a numerical constant $C > 0$ such that whenever*

$$\beta \leqslant Cn^{-1}$$

the following holds. For Lebesgue almost any $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$, there exists $x^* \in \mathbb{S}^{d-1}$ such that the solution $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ to the corresponding Cauchy problem for (SA) (resp. for (USA)) satisfies

$$\lim_{t\to+\infty} x_i(t) = x^*$$

for all $i \in [n]$.

Moreover, when $d = 2$ one can take $\beta \leqslant 1$ and the same conclusion holds.

The proof can be found in Appendix C. As for Theorem 4.1, we briefly discuss the general strategy in Section 5 just below. The improvement to $\beta \leqslant 1$ when $d = 2$ is due to the recent paper [CRMB24].

5. A single cluster for large β

The chain of thought leading to the proof of Theorem 4.3 also leads to

Theorem 5.1. *Fix $d, n \geqslant 2$. There exists a constant $C = C(d) > 0$ depending only on d such that whenever*

$$\beta \geqslant Cn^2,$$

the conclusion of Theorem 4.3 holds for both (SA) and (USA).

We refer the interested reader to Appendix B for the proof, which, much like those of Theorem 4.1 and 4.3 relies on the gradient flow interpretation of the dynamics. We give a general outline thereof. Since all the intervening functions and metrics are real-analytic, the celebrated *Łojasiewicz theorem* [Loj63] implies that for any initial condition, the gradient flow converges to some critical point of E_β as $t \to +\infty$. The genericity over the initial configurations seen in the statements comes from the *center-stable manifold theorem* (Lemma A.1). The former ensures that for almost every initial configuration, the gradient flow does not converge to a strict saddle point of E_β (namely critical points where the Hessian has at least one positive eigenvalue). Whence, generically over the initial configurations, the gradient flow converges to a local maximum. One is then left analyzing the landscape of the underlying energy, with the goal of ensuring that all local maxima are necessarily global.

6. The high-dimensional case

We now elucidate the role that the dimension d plays in clustering results. To start, it turns out that the restrictions on β provided by Theorems 4.3 and 5.1 are specific to the two-dimensional case. The following result is shown in [MTG17, CRMB24].

Theorem 6.1 ([MTG17, CRMB24])**.** *Fix $n \geqslant 2$, $d \geqslant 3$ and $\beta \geqslant 0$. Then the conclusion of Theorem 4.3 holds for both* (SA) *and* (USA)*.*

In Section 6.1 we show that the convergence of Theorem 6.1 is exponentially fast when $d \geqslant n$ (although, with a decay constant that is exponentially small in β) and in Section 6.2 we describe the full dynamics when n is fixed and $d \to +\infty$. We discuss some numerical experiments and posit questions on the intermediate behavior of the dynamics when $\beta \gg 1$ in Section 6.3.

We were made aware of Theorem 6.1 after the first version of this manuscript. The downside of our original proof, which caused us to miss the full range of β, is that we focused on $d = 2$, where the natural perturbations to a critical point involve selecting which particles are moving and which are standing still. However, in $d > 2$, one can move all particles in the same direction (as is done in the proof in [CRMB24]). Using that there is a continuum of directions and only finitely many points, one can find some direction that brings all of them closer.

Remark 6.2 (Invariant measures)**.** *In the theory of dynamical systems, often the first question of interest is to find smooth invariant measures. It is clear that whenever conclusions of Theorem 4.3 hold (in particular, for all β, n when $d \geqslant 3$), neither* (SA) *nor* (USA) *may possess a smooth invariant measure.*

6.1. **Clustering at an exponential rate when $d \geqslant n$.** One can ask whether for almost every initial configuration, the convergence provided by all of the results above holds with some rate. The answer is affirmative—and the rate is in fact exponential—when the initial configuration lies in an open hemisphere.

Theorem 6.3. *Let $n \geqslant 1$ and $\beta > 0$. Suppose $d \geqslant n$. Consider the unique solution $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ to the Cauchy problem for* (SA) *or* (USA)*, corresponding to an initial sequence of points $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ distributed uniformly at random. Then almost surely there exists $x^* \in \mathbb{S}^{d-1}$ and constants $C, \lambda > 0$ such that*

$$\|x_i(t) - x^*\| \leqslant Ce^{-\lambda t} \tag{6.1}$$

holds for all $i \in [n]$ and $t \geqslant 0$.

In fact, let Q and K be arbitrary $d \times d$ matrices. Then the same result also holds for the solution to the corresponding Cauchy problem for (2.3) *with $V = I_d$ (or the natural analogue of* (USA) *with these parameters).*

When $d \geqslant n$ and the points $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ are distributed uniformly at random, with probability one there exists[10] $w \in \mathbb{S}^{d-1}$ such that $\langle w, x_i(0)\rangle > 0$ for any $i \in [n]$. In other words, all of the initial points lie in an open hemisphere almost surely. The proof of Theorem 6.3 thus follows as a direct corollary of the following result, which holds for any $n \geqslant 1$ and $d \geqslant 2$:

Lemma 6.4 (Cone collapse)**.** *Let $\beta > 0$ and let $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ be such that there exists $w \in \mathbb{S}^{d-1}$ for which $\langle x_i(0), w\rangle > 0$ for any $i \in [n]$. Consider the unique solution $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ to the corresponding Cauchy problem for* (SA) *or* (USA)*. Then there exists $x^* \in \mathbb{S}^{d-1}$ and constants $C, \lambda > 0$ such that*

$$\|x_i(t) - x^*\| \leqslant Ce^{-\lambda t}$$

[10]This weak version of Wendel's theorem (Theorem 6.7) is easy to see directly.